

Marketing Campaign Performance Analysis Report

Introduction

This project analyzes marketing campaign data to evaluate platform performance and identify factors that contribute to successful campaigns. The analysis was conducted using Python, with KPIs and an interactive dashboard developed to support decision-making.

Data Cleaning

The dataset was prepared through several preprocessing steps:

- Removing duplicate records
- Handling missing values
- Standardizing platform names
- Converting columns to appropriate data types
- Creating additional performance metrics

Key Performance Indicators (KPIs)

Conversion Rate (CR)

Measures the percentage of clicks that resulted in conversions.

$$\text{CR} = \text{Conversions} / \text{Clicks}$$

A higher Conversion Rate indicates a more effective campaign.

Cost Per Click (CPC)

Measures the average cost of generating one click.

$$\text{CPC} = \text{Cost} / \text{Clicks}$$

It helps evaluate advertising efficiency.

Cost Per Acquisition (CPA)

Measures the average cost required to generate one conversion.

$$\text{CPA} = \text{Cost} / \text{Conversions}$$

Lower CPA values indicate more cost-effective campaigns.

Campaign Distribution

The following chart illustrates how campaigns are distributed across different marketing platforms:

Score Formula

Score =

0.40 × norm(Conv. Rate)

+0.30 × norm(Conversions)

+0.20 × norm(Efficiency)

-0.10 × norm(CPA)

RANK	PLATFORM	SCORE	CR %	CPA
1	● Google	0.8623	10.35%	\$24.19
2	● Facebook	0.6179	10.47%	\$25.69
3	● Email	0.4257	10.32%	\$24.93
4	● LinkedIn	0.2352	9.61%	\$24.89
5	● Instagram	0.2219	9.99%	\$25.16
6	● Twitter	0.0058	9.21%	\$24.66

Performance Scoring Model

The following formula is used to rank platform performance:

Score =

0.40 × norm(Conv. Rate)

+0.30 × norm(Conversions)

+0.20 × norm(Efficiency)

-0.10 × norm(CPA)

Platform Ranking

Rank	Platform	Score	CR %	CPA
1	Google	0.8623	10.35%	\$24.19

2	Facebook	0.6179	10.47%	\$25.69
3	Email	0.4257	10.32%	\$24.93
4	LinkedIn	0.2352	9.61%	\$24.89
5	Instagram	0.2219	9.99%	\$25.16
6	Twitter	0.0058	9.21%	\$24.66

Key Findings

- Significant performance differences exist across marketing platforms.
- Some platforms generate higher conversions with lower costs.
- Higher spending does not always guarantee better results.
- Trend analysis helps identify strong and weak performance periods.

Conclusion

The analysis provided valuable insights into campaign effectiveness and platform performance. The integration of KPIs, visualization, and scoring model allows for better data-driven marketing decisions.